

Atlantic bottlenose dolphin



Element A

**Presentation and Justification of the
Problem**

Enrichment Toys





Problem definition & elaboration

2. Problem definition is creating an enrichment toy for the dolphins because we want them healthy mentally and physically. Dolphins could become aggressive to other dolphins or trainers if they become bored. In the wild dolphins are playful and social. According to Ocean Conservancy, dolphins in the wild can swim up to 100 miles per day.



Primary stakeholders

4. Concern of primary stakeholders. Justification of the problem highlights the concerns of

- Indianapolis Zoo
- Dolphins

Secondary Stakeholders

- Public
- Other zoos or aquariums
- Marine Biologists



Research shows

5. Problem justification is based on why dolphins need enrichment toys.

- Through a range of stimuli, enrichment is used to enhance to the quality of the environment the animals reside in and encourage them to demonstrate their species-specific behaviour, ensuring they are mentally stimulated and in optimal health, allowing the dolphins to be in the best mood.

Sources

- [Seaworld Animal Enrichment](#)



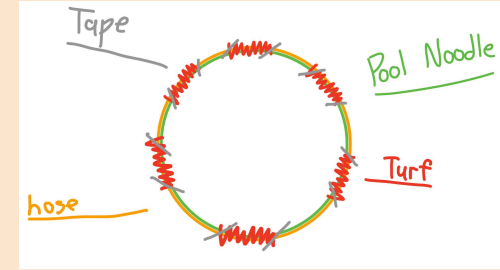
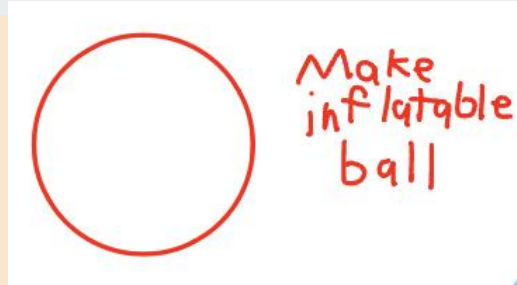
Measurable

7. Determination of design requirements. Design requirements are that it is made without glue or thin rope, it has to be made with materials totaling not over \$100 and can be bought on amazon or at Menards. They also have to be safe with no choking hazards and durable.

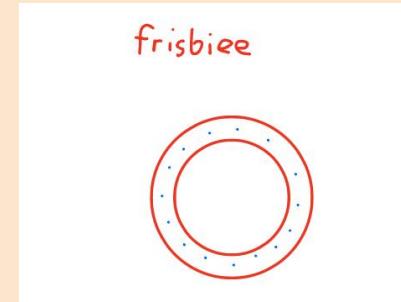
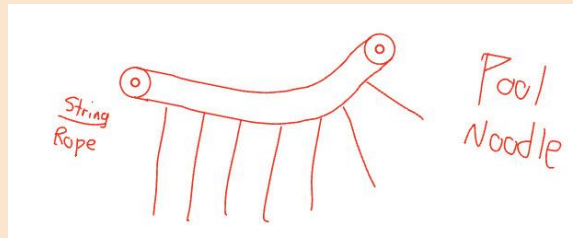
Element B

Documentation and Analysis of Prior Design Attempts

Other designs

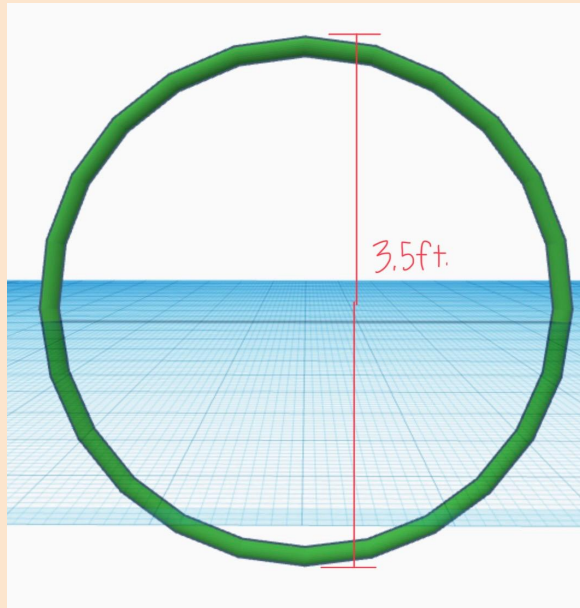


- wild in play with whatever they find like coral, plastic, boats, etc.
- bubbles - trainer or machine
- Zoos will have them paint
- Kong toy
- Pool noodle with string descending (like streamers on a door)
- Beach balls
- pool noodles
- lay on floating mats



Objective & Measurable

2. Our objective is to make a strong dolphins enrichment toy that is big enough to still allow them to swim through and around it.



Element C

Presentation and Justification of Solution Design Requirements



Requirements

7. Determination of design requirements. Design requirements are that it is made without glue or thin rope, it has to be made with materials totaling not over \$100 and can be bought on amazon or at Menards. They also have to be safe with no choking hazards and durable.



Primary Stakeholder Validation

4. Primary Stakeholder Validation. There is evidence that requirements represent the needs of, and have been validated by, the primary stakeholder groups.

Element D

Design Concept Generation, Analysis and Selection

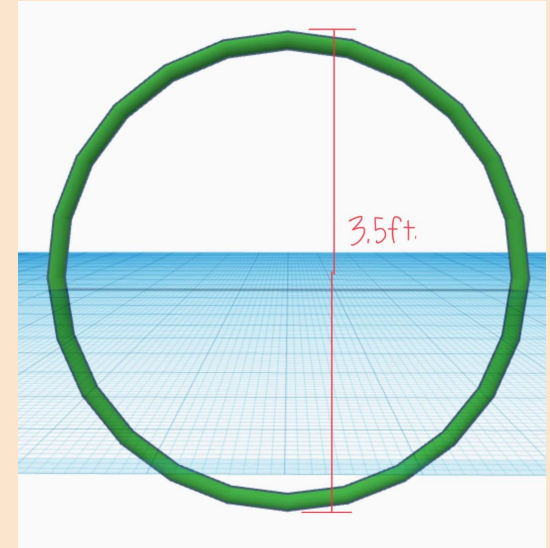
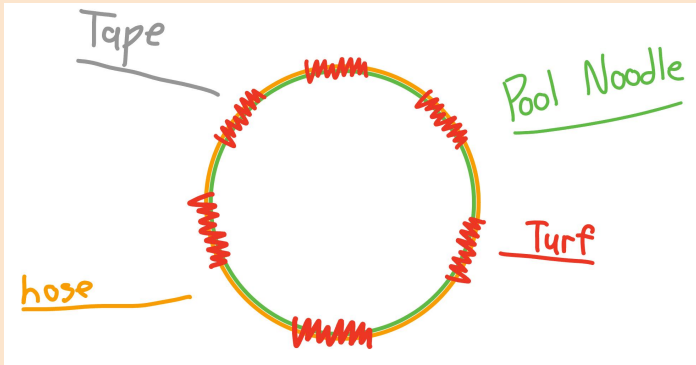


Generating and Comparing Possible Solutions

1. The group wanted the toy to be multi-purpose therefore, we decided to make the toy ring shaped so the Dolphins can swim through it and also covered it with bristles to make it able for the dolphin to scratch itself with it.

Design Solution Choice

2.The design of the toy makes it multi-purpose and able for the dolphin to play with and use to scratch itself





Plan of Action

3. These steps are how to correctly put together and ensure the strength of the toy is strong enough to withstand a dolphins bite.

1st. Connect ends of rubber hose

2nd. Connects ends of pool noodles

3rd. Push rubber hose through pool noodle.

4th. Seal rubber tube in pool noodle.

5th. Wrap in turf.

6th. Seal up hose and noodle.

Element E

Application of STEM Principles and Practices



STEM Principles and Practices to Design Requirements.

1. The STEM Principles and Practices articulated are.

- TinkerCAD, Phone, iPad, computer- technology
- Measurements (size of animal/ size of mouth/budget)→ Math
- Science→ Research Dolphin, materials handle in salt water
- Engineering→ Design, Build, Make improvements



Proposed Solution to STEM Principles and Practices

TinkerCAD→ Design a 3D model of the proposed enrichment toy.

Building the prototype→ making decisions towards the next step

Decision Matrix → help focus/modified a couple of designs

—

Element F



Viable

The difficult/Time consuming part of the project might be building or if there is a critical flaw to the toy and finding a solution

Potential Obstacles we might face is (Building, Materials, and Time Management)

We plan to these obstacles by using time efficiently and effectively to compensate for any time used while solving the problem

How will you know that your end users will be satisfied?

The problems that may be presented are Building, Time Management, and Materials

Document how your methods to addressing those obstacles are supported by end user review

“We may have to _____ because _____ which means _____, which we’ve verified with _____ who told us _____.”

Provide evidence that the obstacles can be addressed in a realistic and sustainable way, supported by end user review ○ “We should be able to address _____ by doing _____, which we confirmed with _____.”

Provide written context to everything presented that provides the reviewer a full understanding of the related work ○ “In order to complete/ensure _____ we will need to _____ which will address _____.”

Element G

Construction Of A Testable Prototype

Final Prototype Iteration Explanation

1. **Final Prototype Iteration Explanation.** The Final Prototype Iteration is a series of pool noodles with silicone tube going down the hole of the noodles for extra strength. The joints are taped together with electrical tape. We cut strips of turf and wrapped the noodle hoop. Wrapping the turf periodically to ensure it stays secure. The purpose of our project was to allow the dolphins an object to swim through or be able to feel texture of the ring.



Final prototype

2. Construction of the Final Prototype. The Final Prototype Iteration is a rubber hose wrapped in pool noodles and turf. The design is held together by electrical tape.





3. Testing or Modeling of Attributes. Several attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are addressed. We plan to test our project in a mini pool to see if it can float.



4.Addressing non-Testable Attributes. Logical justification is provided for those that cannot be tested or modeled mathematically and thus require expert review. Something we took into consideration was the turf possibly falling off. We shook the turf to make sure it didn't fall off.

H. Prototype Testing and Data Collection Plan



Requirements

1. Addressing Requirements. The final prototype Testing Plan addresses requirements of the high priority design requirements.



2. Conduct of Tests. The Testing Plan helps in describing the conduct of tests (through physical and/or mathematical modeling).



Testing

3. Quantity of Tests Planned. The Testing Plan defines how well the enrichment ment toy works of all design requirements based on the instructional context.



Planning

4. Logic of the Plan. The Testing Plan provides details to how we plan on testing and how the use of our enrichment toy will benefit the dolphins.



Field Experts

5. Use of Field Experts. The Testing Plan is confirmed by professor David and Mrs. Soliven.

I: Testing, Data Collection and Analysis



Testing: Quantity

1. Quantity of tests conducted. The student 3 tests for high priority requirements that are reasonable based on instructional context, or through physical or mathematical modeling.



Testing: Quality

2. Quality of tests. The student conducts tests for quality/durability requirements that are reasonable based on instructional context, or through physical or mathematical modeling.



Testing: Comprehension

3. Comprehension of testing procedure. The student demonstrates understanding of testing procedure, including the gathering and analysis of resultant data.



Design: Analysis

4. Detail of the design analysis. The analysis of the effectiveness with which the design met stated goals includes durability, floatation and effectiveness in what it's made for.



Data: Summary

5. Summary of Data Implications. The analysis includes of the data for proceeding with the design and solving the problem.

J: Documentation of Evaluation



Evaluation

1. The standards the project was judged by was it a original design, does it demonstrate the knowledge of the species, does it show evidence of redesigns, does it display planning, challenges, and problem solving, quality of the tri-fold poster, the quality of the 3D constructed prototype and feasibility for implementation .

How the prototype was judged

INDIANAPOLIS ZOO Animal Enrichment Challenge
 April 16th, 2024

School & Team Name: DeCatur Central HS Dolphins

Animal: Red Panda or Dolphin Please circle

	Excellent	Above Average	Average	Needs Work
Original design		✓		
Demonstrates knowledge of species	✓			
Shows evidence of design and redesign	✓			
Displays planning, challenges and solutions	✓			
Tri-fold poster		✓		
3-D Constructed Prototype	✓			
Feasibility for implementation		✓		

Comments:
 "If you had an unlimited budget" was considered thoroughly - further improvement was considered

Evaluator: KAH

INDIANAPOLIS ZOO Animal Enrichment Challenge
 April 16th, 2024

School & Team Name: DeCatur HS - Dolphins

Animal: Red Panda or Dolphin Please circle

	Excellent	Above Average	Average	Needs Work
Original design	✓			
Demonstrates knowledge of species	✓			
Shows evidence of design and redesign	✓			
Displays planning, challenges and solutions	✓			
Tri-fold poster		✓		
3-D Constructed Prototype	✓			
Feasibility for implementation	✓			

Comments:
 love the diff texture of the turf the dolphins would love it

Evaluator: ABIE DAN



Analyzing Evaluation Feedback

3. Yielding a Meaningful Analysis. Documentation synthesizes all relevant categories to yield a meaningful analysis of the external evaluation data.

K: Reflection on the Design Project



Comprehensive Reflection

1. **Clear and Comprehensive Reflection.** The project designer provides clear descriptions on each step of the project



The value of the Judgement was clearly presented and very fair.



What we Learned

Our project idea was very unique and the zoo admired the design

L: Presentation of Designer's Recommendations



Recommendations

1. Clear and Comprehensive Recommendation. The project designer includes recommendations regarding the conduct of a same/similar project in the future.



2. What was recommended was to know how to improve the project when given more time and funds.



3. Detailed Plans for Implementation of Improvements. Plans for Implementation of Improvements are to improve the thickness of the silicone hose and to improve how to make the turf more secure onto the pool noodles.